Diagnosis-Guided Compressed Latent Replay for Continual Document Understanding

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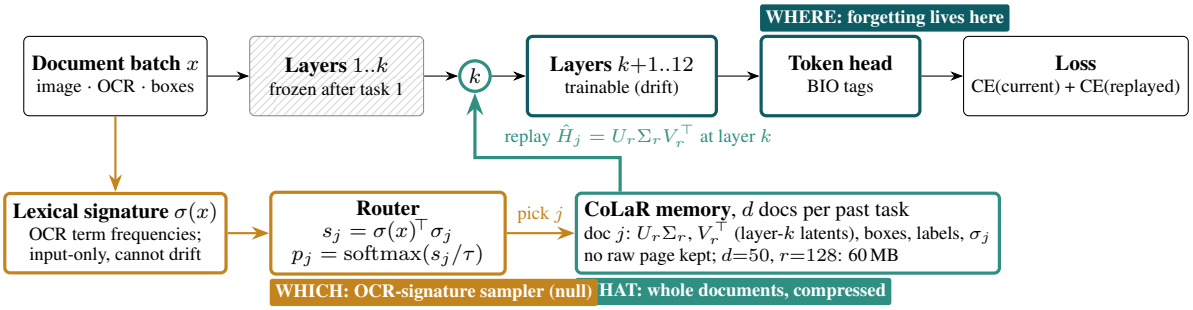


Figure 1. **Diagnosis-guided compressed latent replay (CoLaR).** The anatomy of Sec. 4 fixes *where* to act (head and late layers) and *what* memory must hold (whole documents, Sec. 7): each past document is stored as a truncated SVD of its layer-k activations (60 MB per d=50 task) and rebuilt during replay. A drift-free OCR-signature sampler probes *which* stored document to replay and is reported as a three-seed null (Sec. 7).

Abstract

Continual learning for document understanding is dominated by raw-exemplar replay, which deployed pipelines cannot keep for privacy reasons. We take a diagnosis-first route. Per-group Fisher displacement and layer-wise CKA across four backbones (LayoutLMv3, LiLT, BROS, BERT), three scenarios, and three seeds show that forgetting in continual key-information extraction is output-localised*—concentrated at the classifier head and late layers—and* architecture-general*. Protecting the locus does not remove forgetting; it* relocates *the drift, so a buffer-free fix cannot work: five such families each fail for the same readout-marginal snap. Guided by the diagnosis, we propose* ***CoLaR****: per-document truncated-SVD latent replay that preserves the (feature, position, label) binding every marginal summary destroys (+27 AA single-variable control). CoLaR retains* 87.8±0.5 *AA on DIL at* 60 *MB—within* 1.1 *of the joint oracle, matching raw replay at* 2.7× *less storage—while storing no raw bytes. Under class-incremental label growth the shared softmax, not the memory, is the bottleneck: once each session's test set is scored on its own label set, the oracle rises from* 33 *to* 92 *AA while every replay row stays near* 20–30*. Giving each session its own background logit (****CoLaR-SBR****: a session-blocked readout over the same bank, routed per token with no task id) reaches* 88.0±1.9 *on CIL-CORD and* 89.7 *on the mixed stream, at or above the joint oracle; recalling each session's trunk state through a per-session LoRA delta (SBR-Recall) reaches* 92.7 *with exact retention (single seed). On image CIL the same bank is beaten by a Gaussian-head method whose document analogue is null, isolating whole-document binding as the modality-discriminating ingredient. Code and runs released on acceptance.*

1. Introduction

Organisations process a continuous stream of visually-rich documents—forms, invoices, receipts, contracts—whose meaning depends not only on the words but on where those words sit on the page. Key-information extraction (KIE) from such documents is cast as token-level sequence labelling over a tri-modal input (OCR text, 2-D layout, page image), and multimodal document encoders such as LayoutLMv3 [14], LiLT [37] and BROS [12] define the state of the art. Deployed systems are not static: new document types appear, schemas are extended, and the input distribu-

tion drifts, while retraining on all history is blocked by cost and by privacy constraints on the earlier data. This is continual learning (CL), and its central obstacle is catastrophic forgetting [25].

The CL literature offers a mature toolbox—importance-weighted regularisation [15], distillation from past models [20], exemplar replay [3, 32], task prompts [34, 40, 41] and per-task low-rank adapters [39]—but almost all of it was developed on unimodal image or text classification, and it treats the network as one homogeneous block: a single quadratic penalty across all parameters, whole-network output matching, or a depth-wise freezing cut chosen by hand. A tri-modal document encoder is manifestly *not* homogeneous. It contains a text stream, a visual stream, layout embeddings, shared attention blocks at increasing depth, and a task head, trained on different signals with different inductive biases. The modern parameter-efficient and modular families do not resolve this; they sharpen it, because deciding *which* weights to adapt or *where* to inject a prompt presupposes knowing which part of the encoder is at risk.

Problem and approach. Under continual token-level KIE—a domain-incremental sequence over FUNSD, SROIE and CORD, a class-incremental split of CORD, and a mixed sequence that alternates the two—privacy constraints on earlier documents rule out raw-exemplar replay. The literature's buffer-free alternatives perform below replay by a large margin, and no existing analysis says why in a document setting. We follow a diagnosis-first route. First we localise forgetting inside a document encoder (Sec. 4), then show that the diagnosis explains the remedy ordering, including a backbone-dependent sign flip in parameter-space remedies (Sec. 5). The diagnosis makes a sharp prediction: because the head is where forgetting lives *and* the drift relocates when the head is protected, a buffer-free fix cannot work; we falsify five consolidation families against this prediction (Sec. 6). Finally we read off what a buffer *must* contain, and build it (Sec. 7): per-document SVD-compressed latent replay (**CoLaR**) preserves the (feature, position, label) binding of a document while compressing it, matches raw-exemplar replay at 2.7× less storage, and never leaves raw document bytes in the memory. Where the label space grows, the same diagnosis points at the readout rather than the memory, and a session-blocked readout (**CoLaR-SBR**) closes the class-incremental gap to the oracle (Sec. 8).

Contributions.

1. A multi-backbone, multi-scenario *anatomy* of forgetting in document IE: output-localised, architecture-general, and *conserved* under architectural constraint—protecting the diagnosed locus relocates the drift to whatever remains plastic (Sec. 4 and Fig. 3).
2. A prediction-tested benchmark of six classical strategies over four backbones and three scenarios (68/72 cells, three seeds), with the diagnosis explaining both the global ordering (replay reaches the joint bound; buffer-free methods stay at the naive floor) and a 17-point sign flip in EWC's benefit across backbones (Sec. 5).
3. A convergent negative result over five buffer-free families with a single mechanistic cause—a readout-marginal snap on a class-asymmetric substrate—closed by registered kill-tests (Sec. 6).
4. **CoLaR**: per-document truncated-SVD latent replay guided by the diagnosis. It stores no raw bytes, reaches 87.6 AA at 60 MB on DIL (87.8 over three seeds, within 1.1 of the joint oracle 88.7 and matching raw-exemplar ER at 2.7× less storage), and quantifies whole-document consistency as the necessary buffer ingredient (+27 AA single-variable control). A drift-free OCR-signature sampler is pre-registered and reported as a three-seed null (Sec. 7).
5. **Label growth is a readout problem, and a session-blocked readout solves it.** The "floor regime" on CIL-CORD is largely a scoring artefact (an oracle trained on all labels scores 33 when each session's gold hides the other sessions' entities, 92 when it does not); under session-restricted scoring every shared-softmax method still sits at 20–30 AA, because partial supervision from current and replayed documents contradicts on the one background logit. **CoLaR-SBR** gives each session a block $[\mathrm{O}_b, \mathrm{tags}_b]$ with its own background, trained only where its annotation is complete, and routes per token by $1 - p_b(\mathrm{O}_b)$ with no task id: 88.0±1.9 on CIL-CORD and 89.7 on Mixed, at the oracle. **SBR-Recall** stores a per-session LoRA delta of the upper trunk next to the bank and reads each block under its own delta: 92.7 with every retention column flat (Sec. 8 and Tab. 5).
6. **Whole-document binding discriminates modalities.** On image CIL (CIFAR-100, ImageNet-R) a Gaussian-head method beats the same latent bank, and a ridge decomposition shows the trunk is intact and the loss is readout coverage—the inversion that isolates whole-document binding as the ingredient the image domain does not need (Tab. 4).

2. Related Work

Continual-learning families. Following Wang et al. [38], remedies fall into regularisation (EWC [15], SI [44], MAS [1]; output-side distillation, LwF [20]), replay (GEM [22], ER [32], DER++ [3], iCaRL [31]), optimisation-geometry (GPM [33], OGD [6], C-Flat [2]), representation-based methods over a frozen backbone (L2P [41], DualPrompt [40], CODA-Prompt [34], O-LoRA [39], CL-LoRA [11], RanPAC [26]), and architecture-based isolation [23, 43]. Replay is empirically the strongest fam-

ily [24, 38] but stores raw data, which document deployments cannot keep. The buffer-free alternatives we test in Sec. 6 are drawn from the strongest current members of each line: model merging with classifier realignment (LCA [35]), subspace transfer, and exemplar-free *feature* replay—class-conditional Gaussians in a frozen feature space (PASS, FeTrIL, FeCAM [8, 29, 52]).

Latent and compressed replay. CoLaR descends from a line that stores mid-network activations instead of raw inputs. REMIND [10] freezes the lower stack and quantises later-layer features with product quantisation. Latent Replay [28] banks activations at a chosen layer and injects them during training, in CORe50-style streams. Brain-inspired generative replay [36] replaces exemplars with a VAE sampled per class. Each stores *one summary per token* and is trained for class labels; CoLaR stores *one SVD per document* and is trained for token-level BIO, so it must keep the binding between a token's feature, its position and its label—which the diagnosis of Sec. 7 shows is exactly what every marginal summary destroys.

Where forgetting lives. Ramasesh et al. [30] localise forgetting in split image-classification tasks by resetting or freezing blocks of layers and find it concentrated in the deeper layers nearest the output; Zhai et al. [45] show the same picture carries over to multimodal LLMs used as classifiers. Multimodal CL benchmarks add hints that architecture shapes forgetting—dual-stream models forget less than single-stream ones and deeper layers forget more [48]—and per-layer budget routing (D-MoLE [7]) implicitly assumes non-uniform need without measuring it. None of these attributes forgetting to the components of a single tri-modal encoder, tests whether the location is architecture-invariant, or asks what happens to the drift when the located component is protected. We take the checkpoint-pair, per-layer stance of Ramasesh et al. [30] and extend it along two axes—component and backbone—and add the protection/migration experiment.

Continual information extraction. Continual NER confronts the semantic shift of the "O" tag with distillation and pseudo-labelling (CPFD [46], CFNER [50], ICE [21], SpanKL [51]) or by generating spans (GenCNER [42]); continual RE pairs episodic memory with relation prototypes (EMAR [9], RP-CRE [5], CRL [49]) or uses prompt pools under task-incremental assumptions (WAVE-CRE [18]). All are sentence-level and text-only. Document-level CL is nascent: GDI-Bench [19] shows an MLLM forgets under fine-tuning and mitigates it by *heuristic* layer freezing; continual table detection [27] uses plain ER; ProtoNER [17] reaches LayoutLM-family encoders only in a few-shot regime. Continual fine-tuning of document MLLMs (e.g. InternVL, DocLLM) is an adjacent problem at a different scale; we scope this study to encoder-only backbones where the per-component diagnosis is tractable. No prior work diagnoses which component of a document encoder forgets, and none evaluates the classical CL families on multimodal KIE under one protocol across backbones.

3. Setup

Task. A document is a triple $D = (W, B, I)$ of OCR tokens, normalised bounding boxes and the page image; the model f_θ emits BIO labels per token and is trained with token-level cross-entropy. Performance is entity-level F1 with BIO-aware span matching (`seqeval`). In the continual setting the learner sees tasks $\mathcal{T}_1, \ldots, \mathcal{T}_T$ and at task t may access only $\mathcal{D}_t$ and, for replay methods, a bounded memory. OCR and boxes are dataset-provided throughout, so no OCR engine enters the loop.

Datasets and scenarios. We use FUNSD (149/50 train/test forms, 7 BIO tags), CORD (800/100/100 receipts, 30 fine-grained classes, 61 tags) and SROIE (626/347 receipts, 9 tags). **DIL**: FUNSD→SROIE→CORD under a fixed unified schema {HEADER, KEY, VALUE, OTHER}; the label space is constant and only the input distribution shifts, so forgetting is pure representation drift. **CIL-CORD**: CORD's 30 classes introduced six at a time over five sessions in a fixed easy-to-hard curriculum; the head grows, and tokens of not-yet-introduced classes are absorbed into "O", whose meaning therefore shifts. **Mixed**: a six-session sequence alternating new-class transitions within CORD with new-domain transitions to FUNSD and SROIE, ending in a FUNSD revisit. Session details and the DIL mapping are in the supplementary material (Sec. A).

Backbones. LayoutLMv3-base [14] is the primary tri-modal encoder (text, layout, image; single-stream). LiLT-base [37] and BROS-base [12] are layout-aware but vision-free; BERT-base is a text-only external comparator. Every method runs unchanged on all four through a shared wrapper contract (head expansion, parameter groups, gradient checkpointing); heads are plain linear layers.

Methods. Every comparison uses one frozen reference set of twelve rows: two bounds—naive sequential fine-tuning (lower) and joint multi-task training on the pooled data (oracle upper)—and ten published baselines, one per family and era. They are ER [32] and DER++ [3] for replay (buffer 200 documents or 2,000 images; DER++ α=β=0.5), LwF [20] for distillation (T=2), CODA-Prompt [34] for prompts, CL-LoRA [11] for parameter-efficient adaptation (rank 8), LCA [35] for model merging, and two *modality-paired* slots holding the strongest head-side and representation-side

method of each modality: SLCA [47] and RanPAC [26] on image class-incremental learning, CPFD [46] and IS3 [4] on document continual learning. Dominated relatives (EWC, L2P, DualPrompt, O-LoRA, ER+C-Flat++) are not part of the comparison set; EWC remains in the generality grid (Tab. 1) only as a regularisation reference, and the others appear in the supplementary material (Sec. B). The falsified buffer-free families of §6 and the compressed-replay controls of §7 run on LayoutLMv3/DIL unless stated.

Protocol. AdamW, learning rate 5×10^{-5}, weight decay 0.01, gradient clipping 1.0. Each task trains *to convergence*: early stopping on the task's validation entity-F1 (patience 2, δ=0.1) under a 100-epoch ceiling that never binds, restoring the best-validation checkpoint at each boundary. FUNSD and SROIE have no validation split; for them the stopping signal is read from the evaluation set, a mild optimistic bias shared by every method. Grid results are mean $\pm$ s.d. over seeds $\{42, 7, 123\}$. *Session-restricted scoring.* In CIL-CORD and the CIL halves of Mixed, session j's test gold hides every other session's entity as "O", so a model that correctly labels another session's entity on the same receipt is charged a false positive. Tabs. 1 and 3 use this legacy scoring; Tab. 5 restricts predictions to session j's label set when computing $R[t, j]$ (a no-op on DIL), which is the scoring the CIL literature's per-task matrices assume.

Metrics and the localisation protocol. From $R[i, j]$ (F1 on task j after training task i) we report AA $= \frac{1}{T}\sum_j R[T, j]$ and BWT $= \frac{1}{T-1}\sum_{j<T}(R[T, j]-R[j, j])$; forward transfer is uninformative under label growth (zero-shot term ≈ -85). Localisation is one backbone-agnostic protocol (released as a library call) over a checkpoint pair $(\theta^{(t-1)}, \theta^{(t)})$, a partition of the parameters into groups g (by component, or by depth read off the module names), and an old-task probe set. *Localise*: per-group old-task-Fisher-weighted displacement $D_g = \sum_{\phi\in g} F_\phi^{(t-1)}(\theta_\phi^{(t)} - \theta_\phi^{(t-1)})^2$, large exactly where parameters that mattered for the previous task were overwritten, and per-token linear CKA [16] between the checkpoints at one depth-matched probe per group. *Test*: dominant group vs. each rival (Mann–Whitney over seeds, Bonferroni) and, across backbones, a permutation test on the L1-normalised D_g profile for a shared *location*. *Perturb*: protect the located group, retrain, re-run; the anatomy predicts the profile relocates rather than shrinks. D_g is distinct from Fisher *importance* F_g, which peaks at the head for any token classifier by construction (0.22 vs. 0.005 for the next group) and is never read as forgetting.

4. Anatomy of Forgetting

Forgetting is total within one boundary, on every backbone. Under naive sequential fine-tuning the diagonal of R is strong and the sub-diagonal collapses to 1–3 F1: LayoutLMv3 loses FUNSD from 89.1 to 3.3 after one further task, LiLT 83.9→2.1, BROS 88.9→6.6. On DIL the naive floor is 41.3/40.1/40.3/39.4 on LayoutLMv3/LiLT/BROS/BERT against the joint oracle 88.7/83.8/87.5/77.2; on Mixed the joint-to-naive gap is +33.7/ + 31.1/ + 31.4/ + 31.6, a 2.5-point spread (Table 1).

Finding 1: output-localised. Two instruments agree. Per-token CKA between consecutive checkpoints falls monotonically with depth: 1.00 at the text/layout/patch embeddings, 0.78 over early encoder layers, 0.34 mid, 0.25 at the last layer and 0.16 at the classifier. Old-task-Fisher-weighted displacement D_g (Fig. 2) is one to four orders of magnitude larger at the head than at any encoder bucket, on all four modality-mask conditions of LayoutLMv3, on BERT, LiLT and BROS. A per-component test rejects the null of uniform displacement with the head dominant ($p < 0.01$). The locus is not fusion-specific: the full model (C4), text+layout (C3) and genuine text-only LayoutLMv3 (C1) forget comparably (BWT −81, −88, −79), so removing the visual and layout streams barely changes the profile; forgetting is carried by the text-and-head path common to all three. The text stream is the one that supports usable optimisation at all—with text removed, two of three seeds collapse to an all-background predictor.

Finding 1′: architecture-general. Permuting LiLT's and BROS's per-depth displacement profiles against the LayoutLMv3 reference fails to reject a shared *location* (p = 0.38, 0.51), and BERT reproduces the same head-dominant, depth-monotone pattern. We claim only shared location: at three seeds a cross-backbone comparison of forgetting *magnitude* is underpowered, and we do not read the small differences in AA or BWT between backbones as meaningful.

Finding 2: protecting the locus relocates the drift. If forgetting lives at the head and late layers, the natural intervention is to protect them. On DIL/LayoutLMv3 (seed 42) freezing everything after task 1 gives AA 41.1, BWT −0.0: nothing is forgotten and nothing is learned. Freezing only the diagnosed head and late layers, leaving the early and mid encoder plastic, gives AA 38.8 with BWT −76.8—naive-level forgetting—while early/mid CKA, which is $\approx$ 1.0 under naive training, collapses to 0.19. The drift did not disappear; it migrated to whatever remained plastic. A control that adds a memory to the frozen locus without protecting it (38.9, −75.6) is indistinguishable, so drift, not the absence of a memory, is the bottleneck. This is why uniform protection fails too: a depth-scaled consolidation penalty spread equally across every bucket collapses to 42.6 AA (it cannot fit the tasks), whereas the same budget concentrated on the

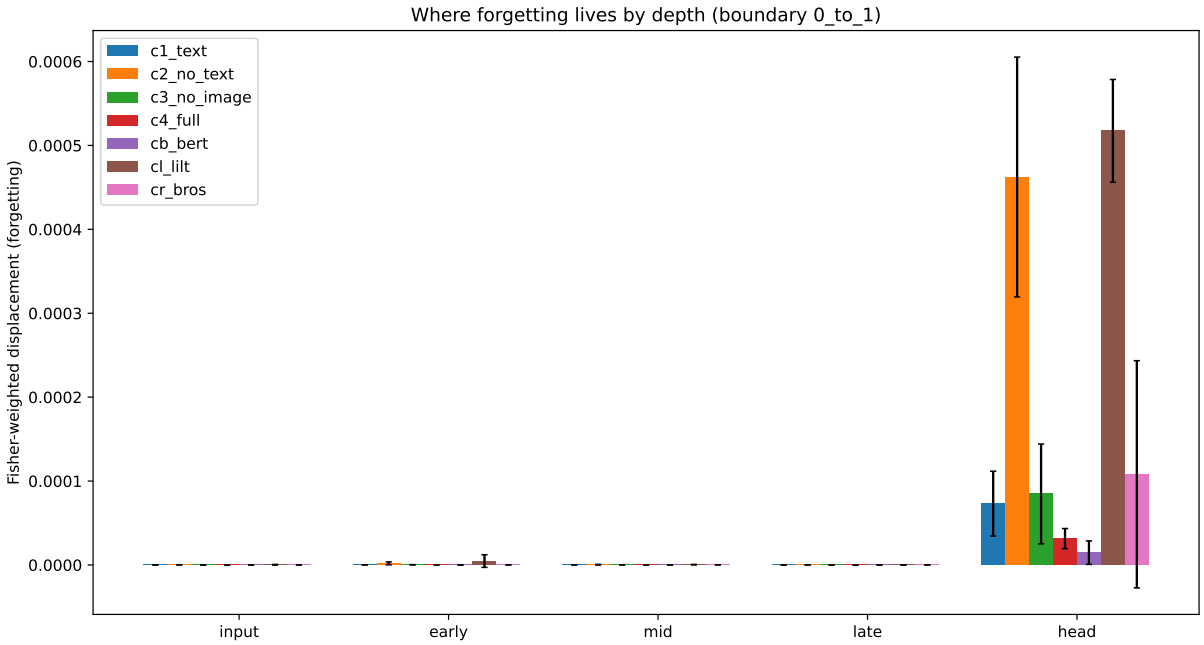


Figure 2. Where forgetting lives. Old-task-Fisher-weighted displacement D_g by depth bucket (log scale, first boundary), for the four LayoutLMv3 modality-mask conditions, text-only BERT, LiLT and BROS; mean over three seeds, ± 1 s.d. The classifier head's weighted displacement is one to four orders of magnitude above any encoder layer's on every backbone.

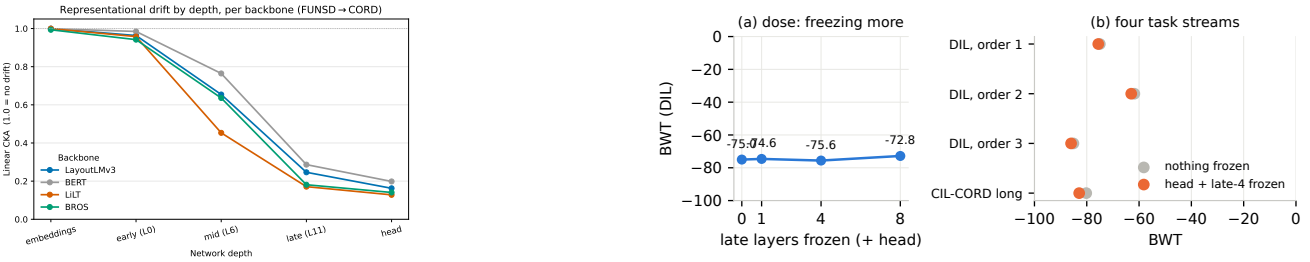


Figure 3. **Forgetting has an address, and protecting it moves it.** Left: linear CKA between checkpoints before and after the second task falls with depth on all four backbones (1.0 = no drift). Right: freezing the head and late layers leaves backward transfer unchanged, (a) whatever the number of frozen layers and (b) on three DIL task orders and a ten-session CIL-CORD stream (LayoutLMv3, seed 42, no memory). The drift relocates to the layers left plastic.

head/late locus recovers 84.7, and head-only (86.1) or late-only (86.2) each recover essentially all of it (three seeds). We read this as a conservation statement: under architectural constraint the total forgetting is redistributed, not removed. Finding 2 is established on DIL with LayoutLMv3 (seed 42 for the freeze arms, three seeds for the consolidation probe); Fig. 3 extends the freeze arm to three task orders and a long class-incremental stream.

Prediction. The anatomy is output-facing and conserved under constraint. It therefore predicts that (i) only a mechanism that re-grounds the head on old-task targets can recover competence, (ii) parameter-space protection—however placed—cannot, and (iii) the ordering should transfer across backbones because the locus does. The next three sections test each.

5. The Anatomy Predicts the Remedy Ordering

Table 1 reports the six classical strategies on four backbones and three scenarios, three seeds each; 68 of 72 cells are complete, and the 4 missing cells are marked rather than imputed.[1]

[1]The missing cells (LwF on LiLT; EWC on LiLT/CIL-CORD) exceeded the memory of the 6 GB GPU that ran the secondary-backbone grid, even with mixed precision; they are reported as missing rather than imputed.

Table 1. Generality grid: average accuracy (mean ± s.d., three seeds) of the six classical strategies on four backbones and three scenarios. *Indep.* is the per-dataset upper bound (one model trained independently on each dataset, three-seed macro over FUNSD/SROIE/CORD). *Joint* is the pooled-data oracle (one model, all three datasets). Independent is only meaningful on DIL, where the task stream is three distinct datasets; CIL-CORD is one dataset split by label and Mixed is a composed stream, so Indep. is undefined there and marked N/A. Grey cells did not fit the available GPU memory (see text). Replay reaches the joint bound on DIL on every backbone; regularisation and distillation stay near the naive floor; on CIL-CORD every method, replay included, sits near the floor.

Scenario	Backbone	Naive	Indep.	Joint	EWC	LwF	ER	DER++
DIL	LayoutLMv3	41.3±1.5	88.2±1.0	88.7±1.1	41.2±1.1	43.7±1.1	87.9±1.4	88.0±0.9
	LiLT	40.1±0.5	85.3±0.8	83.8±1.2	48.4±5.9	–	85.7±0.5	84.4±0.3
	BROS	40.3±1.8	88.8±0.1	87.5±1.3	45.0±4.9	35.2±3.1	87.8±0.4	87.6±1.1
	BERT	39.4±0.3	77.3±0.5	77.2±0.4	46.6±3.0	40.5±0.1	77.1±1.0	77.4±0.6
CIL-CORD	LayoutLMv3	19.0±0.2	N/A	33.3±0.3	0.5±0.6	18.8±0.2	16.0±1.4	18.2±0.3
	LiLT	19.1±0.3	N/A	32.4±0.4	–	–	17.9±0.5	18.0±1.0
	BROS	19.2±0.1	N/A	32.9±0.0	17.3±0.8	23.1±1.2	18.3±1.5	17.7±1.1
	BERT	19.3±0.1	N/A	32.4±0.2	17.9±0.3	24.3±0.3	17.1±1.6	19.2±1.9
Mixed	LayoutLMv3	33.9±0.3	N/A	67.5±0.8	22.0±9.4	35.2±1.2	60.3±5.3	60.8±2.7
	LiLT	30.7±0.7	N/A	61.8±0.6	28.1±3.0	–	57.0±1.4	55.1±2.9
	BROS	33.9±0.4	N/A	65.3±2.1	37.4±3.3	34.4±0.5	61.1±1.4	63.0±0.4
	BERT	23.4±0.3	N/A	55.0±0.6	28.6±0.9	29.4±4.2	50.2±0.6	50.7±1.3

Replay re-grounds the head and reaches the oracle on DIL, on every backbone. ER and DER++ match the joint bound on all four backbones—87.9/88.0 vs. 88.7 on LayoutLMv3, 87.8/87.6 vs. 87.5 on BROS, 77.1/77.4 vs. 77.2 on BERT, 85.7/84.4 vs. 83.8 on LiLT[2]—with BWT of -1 to -3.5. The independent per-dataset upper bound (three models, one per dataset) is 88.2/85.3/88.8/77.3 on LayoutLMv3/LiLT/BROS/BERT—within 0.5 of pooled joint on LayoutLMv3, *above* it on LiLT and BROS, and tied with the replay cells—so the transfer headroom beyond the independent-deployment baseline is sub-point, and the question for a continual learner is whether a single-model solution captures it at a storage cost smaller than 3× model weights. On Mixed, replay closes most of the gap (60.3–63.0 vs. oracles 65–68 on layout-aware backbones; BERT 50.2/50.7 vs. 55.0); the class-incremental transitions inside the sequence are where residual forgetting sits.

Parameter-space protection does not. EWC and LwF remain at the naive floor on DIL for LayoutLMv3 (41.2, 43.7 vs. 41.3). Both mechanisms act on parameters or on whole-network outputs; neither places an old-task gradient on the head, and the anatomy says that is the only place it would matter. EWC's Fisher penalty, computed at the head-dominant locus, does reduce backward transfer (-2.8 vs. -73 for naive on LayoutLMv3) but at the cost of not fitting the new tasks—the same under-fitting mode as the uniform consolidation probe of §4.

[2]Replay slightly exceeds the pooled-data oracle on LiLT; joint on LiLT appears under-fitted by the shared early-stopping rule.

A remedy's benefit is backbone-dependent; the phenomenon is not. Where the naive-to-joint gap agrees to 2.5 points across backbones on Mixed, EWC's benefit over naive spans a 17-point range with a sign flip: -11.8 on LayoutLMv3, -2.6 on LiLT, $+3.5$ on BROS, $+5.3$ on BERT (Fig. 4). LwF likewise helps CIL-CORD on BROS and BERT ($+4$, $+5$) and nothing on LayoutLMv3. A single-backbone comparison of buffer-free remedies can therefore land on either side of zero; a multi-backbone protocol is not optional for this class of claims.

CIL-CORD is a floor regime for everyone. With six fine-grained classes per session over a single receipt domain, every method—replay included—sits at 16–24 AA against an oracle of 33, and on LayoutLMv3 EWC collapses to 0.5. Part of this floor is the scoring: an oracle trained on every label scores 33 because each session's gold hides the other sessions' entities, and 91.7 under session-restricted scoring (Tab. 5). The rest is real and survives the fix: under restricted scoring replay rises only to 20–25 and naive stays at 19, because what must be re-grounded is not a fixed head but a head whose background class is redefined at every session, and replayed documents carry the *old* labelling of the new

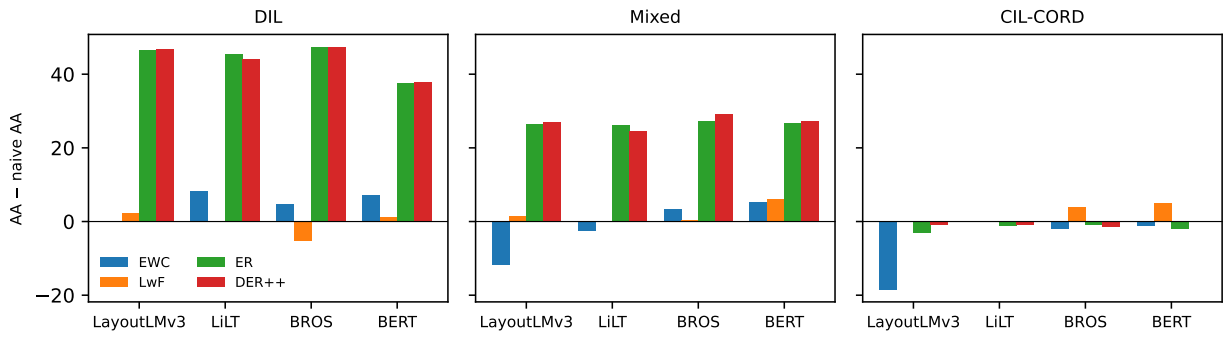


Figure 4. Remedy benefit is backbone-dependent; the phenomenon is not. Change in AA over naive for each classical remedy, per backbone and scenario (mean of three seeds). Replay's benefit is large and consistent; EWC and LwF change sign across backbones (missing bars: cells not run).

classes. Sec. 8 resolves it at the readout.[3]

6. Falsification: Five Buffer-Free Families

Sections 4–5 say replay works because it re-exercises the head on old-task targets. The strongest way to test that reading is to try to get the same effect without a buffer. We built or ported five buffer-free families spanning the mechanisms the literature offers, ran each on DIL/LayoutLMv3, and root-caused each failure (Table 2). None is a contribution; each is a witness.

One mechanism. A per-boundary root-cause analysis localises every failure to the readout. The head's output marginal on old tasks snaps to the just-trained task's gold marginal (cosine 0.997–0.998 for naive and LwF, vs. 0.60 to the old task's own); replay methods stay at 0.99–1.00 to their own. Classes never re-exercised (KEY, HEADER) are extinguished to exactly 0.0 F1 under every modality mask, and the background class collapses too. Forgetting here is a *readout-marginal snap on a class-asymmetric substrate*, not representation erasure—which is why the feature-Gaussian family [8, 29, 52], exercising the head on marginal summaries, only partially recovers (Sec. 7).

Cheap corrections are registered nulls. If the snap were a mis-calibrated prior, correction should reverse it. Five kill-tests registered with explicit bars before any run—marginal matching, prior-ratio correction, per-document EM, a KL anchor to the prior-task mixture, balanced-softmax logit adjustment—recover at best -0.1 AA and leave the extinguished classes at 0.0 F1. A saturation probe explains why: old-class probability mass is $\sim 10^{-6}$, two to three orders below the branch threshold; the information is destroyed before any correction can act. Combined with the migration result of Sec. 4, buffer-free fixes at the head alone are ruled out, and the search shifts to *what minimal object a buffer must store*.

[3]This is the setting in which the continual-NER literature's distillation/pseudo-labelling protocols are designed to operate; we deliberately do not engineer the O-shift away.

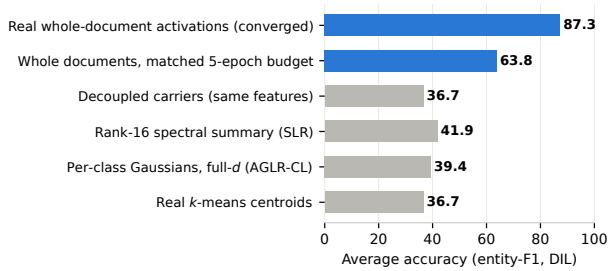


Figure 5. **What the memory must hold** (DIL, LayoutLMv3, layer k=8). Whole documents retain; summaries of the same store fall to the naive level (41.3), and the identical features on decoupled layout carriers lose 27 points against matched whole documents.

7. CoLaR: What a Document Memory Must Hold

Replay works because it re-exercises the head on inputs that keep their old joint labelling; the diagnosis of Secs. 4 and 6 says any admissible memory has to preserve that. The question is what the minimum stored object is. We answer with a falsification ladder over *latent* replay—activations banked at a frozen encoder boundary (layer k) and replayed through the plastic upper layers and head, so storage content varies while the mechanism is held fixed—and read off a memory that keeps every token's feature at its own position with its own label while compressing each document on its own (DIL/LayoutLMv3; single-seed diagnostic runs except where noted).

Table 2. Falsification ledger, DIL/LayoutLMv3 (naive floor 41.3, replay ≈ 88). Each family is the strongest instance we could build of its mechanism; the failure reason is from a per-boundary root-cause analysis, not inferred from the score. †run at a frozen-backbone, short-schedule (3–5 epoch) operating point whose own naive floor is given in the cell. ‡CIL-CORD and Mixed, LayoutLMv3, three seeds each; oracle = true-label refresh ceiling.

Family	Instance	Best AA	Why it fails
Weight merging	TIES/Fisher head merge†	21.6 (no-merge 21.9)	merge of head vectors cancels; near no-op
Merge + realignment	LCA [35] (ported)	43.9±2.5	backbone merge halves BWT; Gaussian realignment kills acquisition
Subspace transfer	HGT / CUBER (ported)†	39.8 (own naive 38.7); HGT ≤ 25.3	protected subspace grows; head unconstrained; α is a pure stability–plasticity dial
Parametric slots	LexSlot (no buffer)	42.2	isolated head slots cannot re-ground shared head
Input-anchored memory	Ledger†	FUNSD 21 / SROIE 23	own-vs-cumulative gap is training-free
Feature-Gaussian replay	LexMem v3b (positive control)	66.0	partial: recovers dense-label tasks only
	LexMem v5 (+relational)	63.2; mid-task 13.0	marginal summary breaks doc consistency
Label refresh (latent memory)	CoLaR-R pseudo (CIL-CORD / Mixed, 3 seeds each)‡	20.0 / 59.5 (plain 20.4 / 58.8; oracle 26.3 / 64.4)	teacher confident on ≈ 0 stale-O tokens; pseudo refresh is a no-op on both scenarios (oracle refresh +5.9 / +5.6 shows the slot is reachable only with true labels)

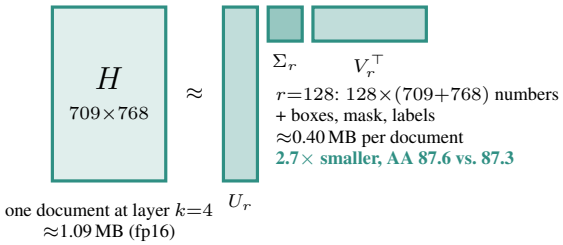


Figure 6. **CoLaR stores whole documents, compressed one at a time.** The pooled layer-k feature space is near full-rank (rank 16 keeps 43% of the variance), but a single document's activations are low-rank: its tokens share one layout and one vocabulary. Truncated SVD per document keeps every token's feature at its own position with its own label, which is the binding that every marginal summary destroys (Fig. 5).

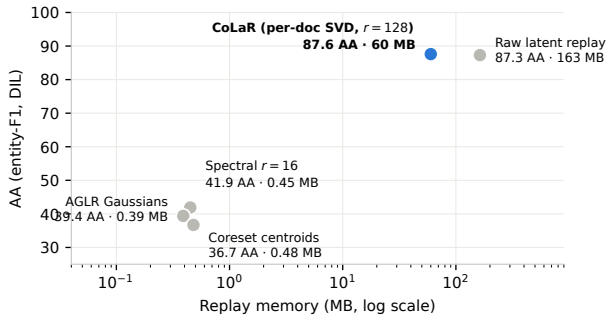


Figure 7. Replay memory versus retention (DIL, log-scale memory). Per-document SVD replay (CoLaR) matches raw latent replay at 2.7× less storage because it preserves whole-document binding; every aggressive summary at ∼0.4 MB breaks that binding and collapses to the naive level.

Whole documents retain; every summary collapses. Replaying the real banked activations of 50 whole documents per task retains AA 87.3 (k=4). Summaries of the *same* store collapse: a rank-16 spectral summary 41.9, per-class Gaussians 39.4, even real k-means centroids 36.7–39.7—so the failure is not synthesis fidelity. The single-variable control isolates the ingredient: four whole replayed documents retain 63.8 at a matched five-epoch budget, while the identical features on decoupled layout carriers retain 36.7. Whole-document binding is worth +27 AA by itself.

The ingredient is domain-discriminating. Tab. 4 runs the frozen reference set on ViT-B/16 image CIL (CIFAR-100 and ImageNet-R, ten sessions, three seeds, buffer 2,000), and the ranking inverts. The buffer-free Gaussian-head method SLCA [47] reaches 87.5/65.0 (joint 89.5/77.6), above raw replay (ER 76.2/54.8, DER++ 76.5/52.8), above merging with Gaussian realignment (LCA 77.2 on CIFAR-100) and above an uncompressed latent bank of 500 images per class (80.0/55.1); it collapses to 38.0/25.8 without its classifier alignment. The image trunk is undamaged: a ridge readout refit on the trained latent-bank trunk reaches 88.3/72.3 (seed 42), above frozen-trunk RanPAC [26] (85.3/58.5) and above SLCA itself, so only the readout statistics are stale. The same probe fitted on the stored bank alone gains only 2 points (80.8/58.0): the headroom is coverage of images the bank does not hold (7.5/14.3 points), not the head's functional form, and the adapted trunk is worth 2.9/15.0 points over the frozen one under the same readout (full decomposition, BWT and the dominated image rows in the supplementary material, Sec. B). CoLaR's per-image SVD at r=128 is lossless against the bank it compresses (79.5 vs. 80.0 on CIFAR-100, 55.3 vs. 55.1 on ImageNet-R) and inherits the same plastic-head gap. On documents the same idea is null in every form (feature Gaussians as a store 39.4; Gaussian realignment after merging 43.9; Gaussian head alignment on CoLaR 86.2 vs. 86.4) while whole-document replay matches joint. An image carries one label, so a class-conditional Gaussian summarises its readout without loss; a document carries one label per token bound to a position and layout, and no per-class statistic of pooled features keeps that binding.

CoLaR, the proposed memory. The two negatives above—every summary of a store collapses while whole documents retain—say the admissible compressor must keep each document's binding. Individual documents are low-rank even though the pooled feature space is not: one document's tokens share one layout and one vocabulary, so its layer-k activation matrix $H \in \mathbb{R}^{N\times d}$ has fast singular-value decay. We therefore **truncate the SVD per docu-**

Table 3. **Fixed comparison set on LayoutLMv3** (AA, mean ± s.d. over seeds 42/7/123): the two bounds and the ten published baselines of the frozen reference set (document-side paired slots CPFD and IS3), then our rows. CoLaR, CoLaR-R and the baselines use 10 training epochs per task with early stopping; LexSlot is the 200-exemplar slot-memory recipe. *did not converge on these sessions, which naive (19.0) and joint (33.3) do learn (supplement, "Null cells among the fixed baselines"). "–": not applicable; IS3 on DIL is a single seed (fixed-label port limitation, supplement).

Method	CIL-CORD	DIL	Mixed
Naive (lower bound)	18.98 ± 0.17	41.31 ± 1.54	33.85 ± 0.33
LwF	18.82 ± 0.15	43.66 ± 1.05	35.23 ± 1.17
ER	15.98 ± 1.36	87.85 ± 1.42	60.32 ± 5.25
DER++	18.15 ± 0.30	**88.02 ± 0.88**	**60.82 ± 2.74**
CODA-Prompt	null*	32.51 ± 1.23	8.11 ± 1.52
CL-LoRA (2025)	15.17 ± 3.14	41.14 ± 0.28	31.37 ± 0.63
CPFD (2023)	null*	6.15 ± 7.29	null*
IS3 (2024)	null*	38.29 (1 seed)	30.10 ± 1.18
LCA (2026)	8.15 ± 2.59	43.87 ± 2.45	19.01 ± 3.68
Text-only comparator (no layout or image):			
BERT-base	19.26 ± 0.11	39.42 ± 0.26	23.37 ± 0.27
CoLaR (ours)	20.38 ± 1.34	87.75 ± 0.48	58.75 ± 1.10
CoLaR-R, sliced (ours)	25.38 ± 2.38	–	61.30 ± 1.91
LexSlot (ours)	19.28 ± 0.12	87.33 ± 1.03	35.49 ± 1.39
Joint (upper bound)	33.32 ± 0.33	88.73 ± 1.13	67.51 ± 0.82
Independent (one model per task, task-ID oracle)	–	88.79 ± 0.55	–

Table 4. **Fixed comparison set on image CIL** (ViT-B/16, ten sessions, AA, mean ± s.d., seeds 42/7/123): same bounds and baselines as Tab. 3 with image-side paired slots SLCA and RanPAC. ER/DER++/SLCA keep 2,000 exemplars; the latent bank and CoLaR keep 500/class. Ridge is a diagnostic readout refit, not a method.

Method	CIFAR-100	ImageNet-R
Naive (lower bound)	11.7 ± 1.1	9.3 ± 0.3
LwF	15.7 ± 0.6	12.7 ± 1.4
ER (2,000 exemplars)	76.2 ± 1.2	54.8 ± 0.7
DER++ (2,000 exemplars)	76.5 ± 1.5	52.8 ± 1.1
CODA-Prompt	15.2 ± 1.8	8.4 ± 0.7
CL-LoRA (2025)	14.1 ± 1.2	11.4 ± 0.8
SLCA (2023)	87.5 ± 0.2	65.0 ± 0.3
RanPAC (2023)	85.3 ± 0.1	58.5 ± 0.3
LCA (2026)	77.2 ± 3.2	54.7 ± 3.6
Latent bank, 500/class (ours, uncompressed)	80.0 ± 0.7	55.1 ± 1.3
CoLaR r=128, 500/class (ours)	79.5 ± 1.0	55.3 ± 1.4
Ridge readout on the latent-bank trunk (probe, seed 42)	88.3	72.3
Joint (upper bound)	89.5 ± 0.4	77.6 ± 0.5

ment: $H_j \approx U_r \Sigma_r V_r^\top$, stored with the document's bounding boxes, attention mask and BIO labels. Replay rebuilds $\hat{H}_j = U_r \Sigma_r V_r^\top$ and injects it at layer k through a forward hook; the loss is CE on the current batch plus CE on the replayed documents. The memory never stores a raw image or token id. Full architecture in Figs. 1 and 6; implementation in `doccl/methods/colar.py` on top of a shared latent-replay base.

CoLaR at r=128 matches raw-exemplar replay. CoLaR retains AA 87.6 (87.8 ± 0.5 over three seeds) at 60 MB, within 1.1 points of the joint oracle, matching raw-exemplar ER (87.9, 200 documents) at a budget that stores no private bytes, and 2.7× smaller than raw latent replay at equal document count (87.3 at 163 MB); r=64 gives 80.6 at 32 MB (Fig. 7). Attempts to shift the memory-side frontier obey a conservation law—class-balanced losses 87.1, coverage-based selection 86.8, entity-weighted compression 87.1 all trade FUNSD for SROIE about 1:1 without moving AA; a three-seed check finds no lever beats the base (balanced 87.8, k-centre 87.0). Read-side extensions (kNN readout blend, metaplastic consolidation) are clean negatives.

Class-incremental documents: live labels help, but the shared softmax is the ceiling. On CIL-CORD under legacy scoring every replay method sits at the naive floor (Tab. 3): CoLaR 20.4 ± 1.3, ER 16.0, DER++ 18.2 against naive 19.0 and joint 33.3. A document banked at session t carries "O" on every entity introduced after t, so replaying it re-teaches the head to suppress exactly the classes it should keep. Refreshing the stored labels from a per-document ledger (the true-label ceiling) gives 26.3 ± 3.6 and a sliced replay loss over the classes alive at banking time gives 25.4 ± 2.4, each above plain CoLaR on every seed (three seeds, pre-registered); teacher-free pseudo-labelling is a registered null (20.0 ± 4.1). Under session-restricted scoring the ordering is unchanged and the gap to the oracle becomes visible: CoLaR 30.4, sliced 42.0, true-label refresh 61.2 (seed 42), joint 91.7. Even with perfect labels the memory recovers a third of the gap, because current documents still label old entities "O" and replayed documents label new ones "O", and both move the same background logit on the same surface forms. The continual-NER comparators built for this failure mode do not converge on these sessions (IS3) and CPFD collapses on every document scenario (Tabs. 3 and 5).

Routing which stored document to replay (null). The OCR vocabulary of a document is set by the input and does not drift with training, which suggests a drift-free sampler (Fig. 1): a softmax over term-frequency signature scores $s_j = \sigma(x)^\top \sigma_j$ in three directions (*near*, *far*, *task*), uniform as the zero-point. The seed-42 reading was promising (far +1.8 over uniform 86.4 at d=50), so we pre-registered that single arm for confirmation. Across three seeds the gap collapses to $+1.8/+0.4/-0.6$, mean +0.5 pp with a sign flip, inside the uniform-sampling spread: lexical routing is a null, and the memory-only headline does not depend on it.

A bounded positive on the storage side. Replaying pseudo-labelled *public* documents through the same mechanism (zero private bytes) recovers dense-label tasks (FUNSD 76.1) but not one the proxy pool does not cover (SROIE ∼ 7), topping out at 58.9: coverage-limited transfer. Budget-

adaptive allocation across tasks moves along a Pareto frontier, not beyond it (supplementary Sec. B).

8. CoLaR-SBR: Un-sharing the Background

Session-blocked readout. The diagnosis above says the contradiction lives in one logit. We therefore un-share it. For each session b that introduces tags the head gains a block $[\mathrm{O}_b, \mathrm{tags}_b]$ with its own background logit, trained with a block-local softmax. A token labelled "O" in session-t data is *truly* O_t, so every session's annotation is complete within its own block. Supervision follows one rule per (document, block) pair: a document trains block b *fully* when its session annotates every tag of b; as *all-background* when b belongs to a different dataset schema, so none of its entities can occur; and not at all otherwise, because its "O" labels are stale for that block. A session that adds no tags (every DIL step, the Mixed revisit) creates no block, so on DIL the method reduces to CoLaR with one block. The trunk is shared, trained by every block, and kept stable by CoLaR's whole-document latent replay; banked documents supervise the blocks of their own session. Inference needs no task id: per block $e_b = 1 - p_b(\mathrm{O}_b)$, the block with the largest e_b claims the token if $e_b > 0.5$, else the token is "O". Closest prior art is ICE [21] (per-session classifiers on a frozen PLM) and SpanKL [51] (per-type sigmoid span heads, no replay); SBR is the blocked readout with a trained trunk and whole-document replay, where each session's data is the full receipt population so every block sees true negatives.

Result. Tab. 5 reports the frozen comparison set under session-restricted scoring. CoLaR-SBR reaches 88.0 ± 1.9 AA on CIL-CORD (final row on seed 42: $91.9/89.1/90.4/89.6/96.0$) against joint 91.7 ± 1.7 and the best baseline, ER, at 24.5 ± 3.4; on Mixed it reaches 89.7 (seed 42) against joint 87.7 ± 1.1 and ER 74.5, with the weakest cell on SROIE (77.2). On DIL, where it collapses to CoLaR with one block, it scores 86.2 ± 1.3. Two ablations locate the effect: the same blocks without the latent bank fall to 20.4 (stale-free labels do not stop representation drift), and the same blocks on a frozen trunk fall to 14.8 (sessions after the first never learn), so the readout and the memory are both necessary and neither is sufficient. Calibrating the block readout further is closed: tuning the threshold and a per-block bias on the test set itself moves AA by at most $+0.6$ on CIL-CORD and $+1.1$ on DIL, so the residual loss is retention, not readout. Storage is 96 MB (bank 50 documents per session at r=128) plus one background logit per block.

SBR-Recall: a memory of brain states. The bank is write-only at test time. SBR-Recall also stores, per session, a LoRA [13] delta of the upper trunk (layers $\geq k$; rank 64 on the attention projections and feed-forward layers) and evaluates block b *under the delta of the session that created it*, so e_b is computed by the network state that learned those tags. The base trunk is frozen after the first session, so old sessions are forgetting-free by construction and a stored delta never goes stale; one forward of the upper layers per distinct delta is needed at inference. The routing is unchanged, task-agnostic and per token, which is what distinguishes this from parameter isolation with a task oracle and from CL-LoRA's shared softmax over per-task adapters (18.6 here despite a diagonal of 93). On CIL-CORD seed 42 SBR-Recall reaches 92.7 AA, final row $95.4/93.2/92.2/89.3/93.3$, BWT $+0.6$, with every column of the retention matrix flat to within a point; it needs no bank at all (54 MB of fp16 deltas in total, 7.1M trainable parameters). At rank 8 on the attention projections alone the same design scores 80.5, with exact retention but sessions two and three plateauing near 65: the isolation is sound and the capacity of the delta is what the look-alike price classes of CORD demand. This is a single seed; two further seeds and the mixed stream are running and the row is marked accordingly.

Table 5. **Session-restricted scoring on LayoutLMv3** (AA, mean $\pm$ s.d.; superscript = seeds when fewer than three, otherwise seeds 42/7/123 for CoLaR-SBR and 42/7 for the baseline rows, whose third seed is in progress). Same frozen reference set as Tab. 3, same runs, re-scored with each session's predictions restricted to its label set. Baseline rows use bf16 autocast, SBR rows fp32. [a]one LwF seed collapsed to all-"O" and is being re-run.

Method	CIL-CORD	Mixed
Naive (lower bound)	19.0±0.1	44.0±0.5
LwF	18.5±0.0	23.9±33.8 [a]
ER	24.5±3.4	74.5±1.6
DER++	20.0±1.8	74.5±0.8
CODA-Prompt	14.9±1.4	11.3±1.1
CL-LoRA (2025)	18.6±0.1	44.8±1.3
CPFD (2023)	1.4±1.9	7.7±10.5
IS3 (2024)	6.6±1.5	20.3±28.7
LCA (2026)	10.3±4.8	38.0 [1]
CoLaR	30.4±2.2	75.6 [1]
CoLaR-R, sliced	42.0±3.1	80.2 [1]
LexSlot	19.0±0.0	50.8 [1]
CoLaR-SBR (ours)	88.0±1.9 [3]	**89.7** [1]
SBR-Recall (ours)	**92.7** [1]	pending
Joint (upper bound)	91.7±1.7	87.7±1.1

9. Limitations

Scope of each claim. Finding 1 (location) rests on four backbones, three seeds and two instruments; cross-backbone magnitude comparisons are underpowered and not claimed.

The benchmark ordering (Sec. 5) is three seeds on 68 of 72 cells, missing cells disclosed, not imputed. Finding 2 (migration) and the buffer-content ladder (Sec. 7) are seed-42 DIL/LayoutLMv3 diagnostics with three-seed checks only at the points stated; CoLaR's headline is three-seed. Of the twelve reference rows, four document cells in three rows are disclosed nulls (Tab. 3); every other cell of the twelve rows is a three-seed measurement (Tab. 4), none imputed. Task order matters (DIL reversed: naive BWT $-81 \rightarrow -75$).

What we do not claim. CoLaR is still a replay method: it needs a store proportional to the number of past tasks. Its hyperparameters (k=4, r=128) were tuned on LayoutLMv3; on BROS (layout-aware) they transfer near-losslessly (87.2 vs. ER 87.8 on DIL), but on BERT and LiLT (text-heavy) the same settings trail raw replay by 5–9 AA points, and per-backbone tuning was not attempted. LiLT's 283M parameters exceeded the 6 GB GPU on CIL-CORD and Mixed even at batch 1 with mixed precision; its separate layout stream is not stored, so latent replay there is approximate. CIL-CORD under legacy scoring is noisy (two runs differing by three relabelled memory tokens differ by 2 AA), so no legacy CIL-CORD claim below three seeds is made. Tab. 5 is a re-scoring of the same runs and is two-seed for the baseline rows until the queue completes; CoLaR-SBR's CIL-CORD row is three-seed, its Mixed row and every SBR-Recall number are single-seed and are claimed only as such. SBR adds one background logit per session and SBR-Recall one upper-trunk forward per session at inference; neither has been run on the secondary backbones. **Early-stopping signal.** Every run early-stops on the current task's evaluation split (best-epoch restoration), the signal all rows share; no row sees any other task's evaluation data during training, and a held-out train split (10%) is being re-run on the headline cells to bound the optimism (Supp.). **Image-side protocol.** The image cells use one house recipe for every row (ViT-B/16 IN-21k, AdamW 5e$-$5, batch 16, resize+flip), which trails the published recipes of SLCA/RanPAC (SGD+cosine, batch 128, IN-1k weights on ImageNet-R) by a uniform offset; absolute numbers are therefore below the literature, the ordering is what we claim. CODA-Prompt and CL-LoRA are run through the shared growing-head protocol *without* the old-class logit masking of their official code; their collapse in Tab. 4 reflects that protocol, and a masked re-run is reported in the supplement.

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